

# On the Dilemma of Reliability or Security in Unmanned Aerial Vehicle Communications Assisted by Energy Harvesting Relaying

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**Abstract**—In this study, we investigate the trade-off between reliability and security in unmanned aerial vehicle (UAV) communications systems, considering a UAV-terrestrial network aided by a relay powered by a dedicated power beacon. For this system, we derive the outage probability (OP) under both exact and approximate frameworks and compute the approximations in the closed-form expressions. For the security aspect, we also derive the intercept probability (IP) under exact and approximated frameworks. To minimize the IP, a friendly jamming technique is employed whereby the power beacon constantly broadcasts artificial noise (AN) toward an eavesdropper. Based on the derived mathematical framework, we then formulate a bi-objective optimization problem by jointly minimizing the OP and IP with respect to the UAV's position and the time-switching (TS) ratio. A suitable algorithm, named non-dominated sorting genetic algorithm version II (NSGA-II), is deployed to obtain the sub-optimal solution. Finally, numerical results are presented to verify the accuracy of the proposed mathematical framework and the superiority of jointly minimizing both the OP and IP using only the channel statistics.

**Index Terms**—Energy harvesting, intercept probability, outage probability, positioning, unmanned aerial vehicle.

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## I. INTRODUCTION

UNMANNED aerial vehicles (UAVs) have gained much attraction in recent times from both academia and industry in fields such as the military, transportation, and surveillance. UAVs are especially flexible and easy to employ and can be deployed almost immediately in disaster areas to establish Internet connections for rescue support and relief activities. UAVs are also able to quickly deliver packets for smart transportation, surveillance, and military targets on battlefields. In wireless communications, UAV-aided networks can improve coverage to remote areas and ameliorate reliability. However, for the true application of UAVs in these fields, many obstacles first need to be overcome, one of these being that because of power constraints in a UAV, it generally cannot operate for long periods or over long distances. As a consequence, other techniques such as relaying networks are often employed in combination with UAV assistance networks. A relaying network is regarded as an effective method of enhancing system reliability because it not only reduces transmission distance, it also compresses background noise with the decode-and-forward (DF) protocol. Nevertheless, the cost of this improvement is an increase in network power consumption. Relays must always be on standby to forward information from the source node, thus reducing the entire network's energy efficiency (EE).

Fortunately, energy harvesting (EH) is an emerging technique suitable for improving energy efficiency in wireless networks. By wirelessly charging the battery of a handheld device, EH addresses the problem of frequently recharging end devices. EH is also a solid step towards genuinely neutral power consumption in future wireless networks. UAV-supported, EH-based relaying networks are therefore a promising solution for the next-generation 5G-Advanced cellular networks. However, if we assess the performance of UAV-supported, EH-based relaying networks in terms of security, they are more vulnerable than conventional point-to-point communications. The basis of this assessment is that eavesdroppers have two attempts to wiretap secure information in dual-hop relaying networks. The eavesdroppers also benefit from the advantages that the UAV brings to a legitimate network. This, therefore, opens the dilemma of whether the relay's assistance is beneficial or detrimental. And can the UAV truly improve reliability? Before highlighting the main

contributions and novelties of the current study, let us review the state of the art in UAV-assisted relaying networks.

The performance of UAV-aided EH relaying networks was investigated extensively in [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], and [15]. In [1], the authors studied outage probability (OP) of UAV-based reconfigurable intelligent surfaces (RIS) assisted wireless networks. Particularly, they derived the OP and bit error rate (BER) of RIS-aided dual-hop UAV communications systems by using a generalized- $K$  distribution to approximate the ground-to-air (G2A) link. However, they did not address the security issue of the considered network. Additionally, the impact of the co-channel interference (CCI) on the system performance was ignored. Furthermore, in order to attain the closed-form expression of the OP, a series of approximations was applied in [1]. The present work, on the other hand, studies both the OP and intercept probability (IP) that takes into consideration energy harvesting techniques and the influences of co-channel interference. The authors in [2] derived the connection probability, secrecy outage probability (SOP), and secrecy rate (SR) of UAV-enabled amplify-and-forward (AF) relaying with simultaneous wireless information and power transfer (SWIPT) and jamming at the destination; they addressed only the performance of these metrics without any further investigation of the effects of some key parameters. Examining cognitive radio networks (CRNs) aided by multiple UAVs under the Nakagami- $m$  channel, the authors in [3] derived the SOP. Although the SOP was studied extensively, it was not minimized and some simulation parameters, for example, carrier frequency and transmission distance, were not explicitly studied. Bao and other authors in [4] also considered UAV-aided relaying systems, deriving the intercept probability and SR. However, they did not employ cooperative jamming or EH to further enhance secrecy performance and energy efficiency. The study in [5] and [6] investigated the OP and IP of the dual-hop AF relaying networks under two EH schemes. The authors nonetheless considered conventional relaying networks in place of UAV communications systems, and although they also considered the trade-off between reliability and security, unlike the current study, the results depended substantially on numerical computations instead of the formulation of a formal problem and mathematical solution. The work in [7], however, used a UAV to improve the flexibility of backhaul networks towards 5G cellular networks. The results of this study demonstrated that with the aid of a UAV, the outage probability in the proposed network was consistently lower than in a conventional backhaul network. Wang et al. in [8] studied the max-min secrecy rate optimization problem. The authors maximized the SR of the worst legitimate link by simultaneously optimizing the UAV's position, the artificial noise (AN) power created by the legitimate receiver at the eavesdropper, and the power splitting ratio of SWIPT. The authors were able to maximize based on an abstract framework rather than an explicitly mathematical SR framework. The optimal number of UAVs and their corresponding positions were analyzed in [9] to ameliorate the signal-to-interference ratio (SIR) of the dual-hop networks under the impact of multiple co-channel interference. The authors mainly focused on improving the system's reliability.

By contrast, the current study considers security in addition to system reliability. Zhu et al. [10] optimized UAV position, the beamforming vector, and power control to maximize the average rate of a dual-hop UAV-assisted wireless network. Again, the authors used an abstract framework as a utility function of the optimization problem instead of an explicit framework. However, security and energy harvesting were not considered. The maximization problem of the UAV relay-aided cognitive radio networks was studied in [11]. Specifically, the authors maximized the average worst-case SR of the secondary networks by concurrently optimizing the UAV's trajectory and transmit power. The work, however, did not study the outage probability of the main link or consider the bi-objective optimization problem, which is addressed in the current study. The maximization of the average secrecy rate (ASR) with respect to the UAV trajectory and power allocation was carried out in [12], [13], and [14]. Nonetheless, they considered different scenarios where the UAV acted as a relay to help exchange secure information in the conventional non-EH-enabled networks while in the present work, we consider the UAV-based EH-enabled networks. Moreover, they studied a single-objective optimization problem (SOOP) while we address the multi-objective optimization problem (MOOP). The work in [15] maximized the secrecy energy efficiency (SEE) of a UAV-aided dual-hop network where the UAV was used to enhance the main link. The study maximized by jointly optimizing the transmit power, communication scheduling, and UAV trajectory. However, as in the work in [8] and [10], an abstract mathematical framework was also used, and a friendly jamming technique to further improve secrecy performance was not applied. The authors in [16] minimized the intercept probability security region which was the area where the IP was less than a predefined threshold under the OP constraint. However, they considered a quite simple networks where a pair of legitimate transmitters and receivers exchanged secure information with the help of a UAV continuously transmitting jamming signals to the eavesdropper. Since they considered a conventional non-EH-enabled networks, the mathematical framework was trivial and the optimization problem as well as the constraints were not the same as in the present manuscript. On the other hand, the works in [17] and [18] studied the minimization problem of the OP. More precisely, authors in [17] minimized the OP by jointly optimizing the power allocation and UAV trajectory. The system model was simple since it comprised the single antenna transmitter, receiver, and UAV to help to exchange information. The work in [18] investigated the OP minimization of the cognitive non-orthogonal multiple access (NOMA) systems by using the machine learning approach. In particular, they minimized the OP of the secondary networks under the constraints of the OP of primary networks and the intercept probability of the eavesdropper. They, nevertheless, count solely on the synthesized data set instead of using popular ones. Additionally, they also did not take into account the energy harvesting aspects.

Apart from the above-mentioned works, the current study investigated the performance of both security and reliability in UAV communications aided by EH-enabled relaying

and cooperative jamming. We derive both the exact and approximate mathematical framework of the OP and IP. We mathematically study the trade-off between security and reliability in the considered networks by employing the explicit closed-form expressions of these metrics. In particular, we formulate a multi-objective optimization problem that jointly minimizes the IP and OP with respect to the UAV's position and time-switching (TS) ratio. The non-dominated sorting genetic algorithm version II (NSGA-II), a multi-objective algorithm, is suggested as a method to find the optimal solution to the problem. The main contributions of the present paper are summarized as follows:

- A proposed UAV communications system aided by an EH-enabled relay which is powered by a beacon that also creates AN signals to decrease the intercept probability of an eavesdropper.
- Transmit antenna selection (TAS) and selection combining (SC) to improve the reliability of the proposed systems. However, the eavesdropper also uses SC to maximize its intercept probability.
- Derivation of the exact and approximate framework of both the OP and IP of the networks, incorporating several correlated random variables (RVs).
- To the best of our knowledge, the joint OP and IP minimization problem is pioneering work. Both metrics are simultaneously minimized with respect to the UAV's position and time-switching ratio.
- Use of the non-dominated sorting genetic algorithm version II to obtain the sub-optimal solution.
- Numerical results using the Monte-Carlo method to verify the accuracy of the derived mathematical frameworks and to gain insight into the considered metrics regarding key parameters.

The remainder of the paper is organized as follows. Section II introduces the system model. Section III presents the framework for analysis of the OP and IP. Section IV studies the joint optimization problem of both OP and IP. Section V presents the Monte Carlo simulations and verifies the accuracy of the proposed framework. Finally, Section VI concludes the paper and presents perspectives.

## II. SYSTEM MODEL

Let us consider a UAV communications system where an unmanned aerial vehicle denoted  $S$  sends information to a ground terminal denoted  $D$  (Fig. 1).<sup>1</sup> The transmission is assisted by an energy-harvesting-enabled terrestrial relay denoted  $R$ . We assume that a direct link between  $S$  and  $D$  does not exist owing to ultra-long transmission and power limitations in the UAV. This system also contains an active eavesdropper, denoted  $E$ , who attempts to intercept information between  $S$  and  $D$ . The relay is not connected to the

<sup>1</sup>The adopted system model can be considered as cellular networks where the UAV acts as an edge base station to serve users in rural areas. However, owing to the obstacles such as a mountain range and/or river, users in these remote areas may not be served by the UAV. Thus an EH-enabled relay is employed to extend the coverage areas of the networks. Here, the EH-based relay is chosen since it is installed in rural areas so connecting to the power grid may not be possible and declines the energy efficiency of the networks.

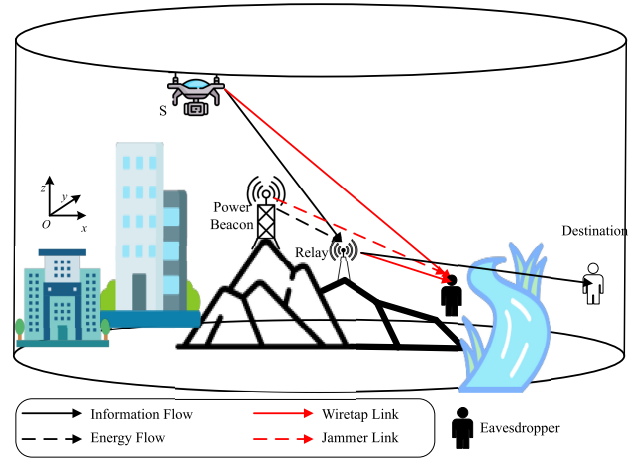


Fig. 1. The proposed UAV-based EH-enabled relaying network.

power grid and counts only on the harvesting energy from the power beacon, denoted  $PB$ . Moreover, since the eavesdropper is classified as active, it is reasonable to employ friendly jamming at the power beacon station. Both  $S$  and  $D$  are equipped with  $M \in \mathbb{N}$  and  $N \in \mathbb{N}$  antennae, whereas  $E$  and  $PB$  are only equipped with a single antenna. Additionally, the global channel state information (CSI) is assumed in the present work. For the legitimate links, it can be obtained via a high-accurate feedback channel and the pilot training approach [19]. Regarding the eavesdropper links, the CSI of the active eavesdropper can be attained via several well-known techniques such as pilot signals [20], energy ratio detectors [21], and minimum description length algorithm [22].<sup>2</sup> Although the global CSI is considered, the negative impact of the imperfect CSI on the OP performance can be effectively mitigated by either increasing the number of transmit and receive antennae as well as the transmit power of the power beacon as shown in Fig. 10 and [26, Fig. 5]. The influences of the imperfect CSI on the IP, on the other hand, is beneficial since it reduces the wiretap probability of the eavesdropper.

### A. Channel Modelling

The transmitted signals are subject to both small-scale fading and large-scale path-loss.

1) *Small-Scale Fading*: Since there is a strong light-of-sight (LOS) path between UAV and all terrestrial nodes, i.e.,  $R$ ,  $E$ , a Rician distribution is therefore adopted to capture this strong LOS component [1]. Let us denote  $h_{S,o}$ ,  $o \in \{R, E\}$  as the channel coefficient from  $S$  to node  $o$ . As a result, the channel gain is followed by non-central chi-square distribution with parameters  $K_{S,o}$  and  $a_{S,o}$ , expressed as follows [27]:

$$F_{|h_{S,o}|^2}(z) = 1 - \sum_{l \geq 0} \sum_{p=0}^l \frac{K_{S,o}^l a_{S,o}^p}{l! p!} z^p \exp(-K_{S,o} - a_{S,o} z), \quad (1)$$

<sup>2</sup>The passive eavesdropper can be detected and its CSI can also be obtained via some techniques, namely, leak oscillator power [23], [24] and/or some sophisticated coding schemes [25]. However, the performance of the considered network under the passive eavesdropper is left for future work.



where  $K_{S,o}$  is the Rician factor and refers to the power of the line-of-sight component to the scattered components and  $a_{S,o} = \frac{K_{S,o}+1}{\lambda_{S,o}}$ ,  $\lambda_{S,o}$ ,  $o \in \{R, E\}$  is the large-scale path-loss (described in Section II-A.2). However, the small-scale fading of transmission links between all terrestrial nodes is followed by a Rayleigh distribution with zero mean and  $\lambda_{q,p}$ ,  $q \in \{PB, R\}$ ,  $p \in \{R, D, E\}$  variance. We assume the block flat fading such that fading remains constant for the entire transmission and changes independently for each transmission.

2) *Large-Scale Path-Loss*: Considering a generic link from node  $q^+$ ,  $q^+ \in \{S, PB, R\}$  to  $p \in \{R, D, E\}$ , denoted  $\lambda_{q^+,p}$ , the large-scale path-loss is then formulated as

$$\lambda_{q^+,p} = L_0 d_{q^+,p}^\delta, \quad (2)$$

where  $L_0 = \left(\frac{4\pi f_c}{c}\right)^2$  is the path-loss constant,  $f_c$  is the carrier frequency (in Hz),  $c = 3 \times 10^8$  (in meters per second) is the speed of light,  $\delta > 2$  is the path-loss constant, and  $d_{q^+,p}$  is the transmission distance. We consider a three-dimensional plane, hence, the position of all nodes is represented by a triplet set, for example, the location of the relay is  $(x_R, y_R, z_R)$ .

### B. Transmission Procedure

The entire transmission occurs in three phases. In the first phase, the EH-enabled relay is charged by harvested energy from the power beacon. The harvested energy is computed from [28]

$$E_R = \eta \alpha T P_B |h_{BR}|^2, \quad (3)$$

where  $\eta \in (0, 1)$  is the conversion efficiency,  $\alpha \in (0, 1)$  is the time-switching factor,  $T$  is the total transmission time.  $|h_{BR}|^2$  is the channel gain from the power beacon to the relay. Without loss of generality, we assume that  $T$  is equal to 1 second, and  $P_B$  is the transmit power of the power beacon. It is noted that in (3), we do not take into consideration the harvested energy from background noise since it is not comparable with the intended signals [29].

In the second phase, the UAV broadcasts information to the relay, and because of the nature of wireless propagation, the eavesdropper also receives this information and attempts to decode it. Meanwhile, to minimize the intercept probability of the eavesdropper, we employ a friendly jamming technique in which the power beacon broadcasts artificial signals that are known to both  $R$  and  $D$  but unknown to  $E$  [8]. The received signals at  $R$  and  $E$  are then computed from

$$\begin{aligned} y_R &= \sqrt{P_S} h_{S,R} x_S + n_R, \\ y_E^2 &= \sqrt{P_S} h_{S,E} x_S + \sqrt{P_B} h_{B,E} x_B + n_E^2, \end{aligned} \quad (4)$$

where  $P_S$  is the transmit power of UAV and  $x_S$  and  $x_B$  are the transmit signals of UAV and PB, respectively. Without loss of generality, we assume that  $\mathbb{E}\{|x_S|^2\} = \mathbb{E}\{|x_B|^2\} = 1$ .  $n_R$  and  $n_E^2$  are the additive white Gaussian noise (AWGN) at the relay and eavesdropper at the second phase.  $h_{S,R}$  is the channel coefficient from the UAV's best antenna to the relay.  $h_{S,E}$  is the channel coefficient from the selected antenna at UAV to an eavesdropper, but it needs not necessarily be the best channel coefficient from all the available links.  $h_{B,E}$  is the

channel coefficient from the power beacon to the eavesdropper. Here, to take advantage of multiple antennae at the UAV, transmit antenna selection is yielded to attain full diversity order. Additionally, TAS is selected instead of the maximal ratio transmission (MRT) technique because it is more suitable for UAVs by being less complex and consumes less power. The antenna selected for the best channel gain from  $S$  to  $R$  is

$$b = \arg \max_{m \in \{1, \dots, M\}} (|h_{S_m R}|^2). \quad (5)$$

In the final phase, the relay amplifies signals from the UAV and forwards them to the ground terminal while the PB continues to broadcast artificial noise to reduce the possibility of snooping by the eavesdropper. The received signals at  $D$  and  $E$  during this phase are given by

$$\begin{aligned} y_{D_c} &= h_{RD_c} \beta y_R + n_{D_c} \\ y_E^3 &= h_{RE} \beta y_R + \sqrt{P_B} h_{BE} x_B + n_E^3, \end{aligned} \quad (6)$$

where  $h_{RD_c}$  is the channel coefficient from the relay to the best antenna at the destination resulting from the selection combining (SC) technique, formulated as  $c = \arg \max_{n \in \{1, \dots, N\}} (|h_{RD_c}|^2)$ .  $h_{RE}$  is the channel coefficient from the relay to the eavesdropper.  $n_{D_c}$  and  $n_E^3$  is the AWGN at the destination and eavesdropper during the third phase.  $\beta$  is the amplification factor selected to provide a constant transmit power at the relay, expressed as

$$\beta = \sqrt{\frac{P_R}{|h_{S_b R}|^2 P_S + N_0}} \approx \sqrt{\frac{P_R}{|h_{S_b R}|^2 P_S}}. \quad (7)$$

Here  $N_0$  is the noise variance of the AWGN at relay, evaluated as  $N_0 = -174 + \text{NF} + 10 \log(\text{Bw})$  [dBm], while  $P_R$  is the transmit power of the relay.  $P_R$  relies on the harvested energy in the first phase and is expressed as

$$P_R = \frac{E_R}{(1 - \alpha) T / 2} = \mu P_B |h_{BR}|^2, \quad (8)$$

where  $\mu = \frac{2\eta\alpha}{1-\alpha}$ . A direct inspection of (8) indicates that the larger the  $\alpha$  the higher the transmit power at the relay, which benefits both the main and eavesdropping links. Substituting (8) and (7) into (6), the signal-to-interference-plus-noise ratio (SINR) at  $D$  and  $E$  (at the third phase) are rewritten as follows:

$$\begin{aligned} \gamma_{D_c} &= \frac{P_S \beta^2 |h_{S_b R}|^2 |h_{RD_c}|^2}{|h_{RD_c}|^2 \beta^2 N_0 + N_0} = \frac{\mu \Phi \Psi |h_{BR}|^2 |h_{S_b R}|^2 |h_{RD_c}|^2}{\left(\mu \Psi |h_{BR}|^2 |h_{RD_c}|^2 + \Phi |h_{S_b R}|^2\right)}, \\ \gamma_E^3 &= \frac{P_S \beta^2 |h_{S_b R}|^2 |h_{RE}|^2}{|h_{RE}|^2 \beta^2 N_0 + P_B |h_{BE}|^2 + N_0} \\ &= \frac{\mu \Psi \Phi |h_{BR}|^2 |h_{S_b R}|^2 |h_{RE}|^2}{\left(\mu \Psi |h_{BR}|^2 |h_{RE}|^2 + \Psi \Phi |h_{BE}|^2 |h_{S_b R}|^2 + \Phi |h_{S_b R}|^2\right)}, \end{aligned} \quad (9)$$

where  $\Psi = \frac{P_B}{N_0}$  and  $\Phi = \frac{P_S}{N_0}$ . Here  $\Psi$  and  $\Phi$  are referred to the average transmit-power-to-noise ratio of the power beacon and the source node. The physical meaning of these parameters is to measure how strengths of the average transmit power relative to the background noise. To maximize the eavesdropping probability, the eavesdropper combines its received



signals in the second and third phases by employing the selection combining technique; the end-to-end (e2e) signal-to-interference-plus-noise ratio is then formulated as follows:

$$\gamma_E = \max(\gamma_E^2, \gamma_E^3) = \max\left(\frac{\Phi|h_{S_bE}|^2}{\Psi|h_{BE}|^2 + 1}, \frac{\mu\Psi\Phi|h_{BR}|^2|h_{S_bR}|^2|h_{RE}|^2}{\mu\Psi|h_{BR}|^2|h_{RE}|^2 + \Psi\Phi|h_{BE}|^2|h_{S_bR}|^2 + \Phi|h_{S_bR}|^2}\right), \quad (10)$$

where  $\gamma_E^2$  is computed from (4) and is given as

$$\gamma_E^2 = \frac{P_S|h_{S_bE}|^2}{P_B|h_{BE}|^2 + N_0} = \frac{\Phi|h_{S_bE}|^2}{\Psi|h_{BE}|^2 + 1}. \quad (11)$$

*Remark 1:* Direct inspecting  $\gamma_{D_c}$  and  $\gamma_E$  in Eqs. (9) and (10), we observe that both SINRs have the common RVs  $|h_{S_bR}|^2$  and  $|h_{BR}|^2$  that are a function of the UAV coordinates. Additionally, both the instantaneous rate of the destination,  $C_{D_c}$ , and eavesdropper,  $C_E$  given in (12), is a function of the time-switching ratio as well. As a consequence, one needs jointly minimize both metrics with respect to the UAV coordinates and time-switching ratio in order to attain the optimal solution. It is noted that if two single-objective optimization problems are considered instead of a multi-objective optimization problem, the solution will be sub-optimal since it does not take into consideration the correlation between these utilities' functions.

The instantaneous rate at the destination and eavesdropper are therefore formulated as

$$C_{D_c} = \frac{(1-\alpha)}{2} \log_2(1 + \gamma_{D_c}),$$

$$C_E = \frac{(1-\alpha)}{2} \log_2(1 + \gamma_E). \quad (12)$$

Inspecting (12), we observe that increasing  $\alpha$  will obviously decrease the capacity of both the main and eavesdropper links; this is in contrast to the impact on the transmit power of  $P_R$ . As a consequence, an optimal value of  $\alpha$  will always exist yet compromise the security and reliability of the considered networks.

### III. PERFORMANCE ANALYSIS

The current study investigates the trade-off between reliability and security for UAV-terrestrial systems. We examine particularly the outage probability performance, which represents the system's overall reliability. Security, however, is measured by the intercept probability at the eavesdropper. Mathematically speaking, these metrics are formulated as follows:

$$\begin{aligned} \text{OP} &= \Pr\left(C_{D_c} = \frac{(1-\alpha)}{2} \log_2(1 + \gamma_{D_c}) \leq R\right) \\ &= \Pr\left(\gamma_{D_c} < \gamma_{th} = 2^{\frac{2R}{(1-\alpha)}} - 1\right) \\ &= \Pr\left(\frac{\mu\Psi\Phi|h_{BR}|^2|h_{S_bR}|^2|h_{RD_c}|^2}{\mu\Psi|h_{BR}|^2|h_{RD_c}|^2 + \Phi|h_{S_bR}|^2} < \gamma_{th}\right), \\ \text{IP} &= \Pr\left(C_E = \frac{(1-\alpha)}{2} \log_2(1 + \gamma_E) \geq R\right) \end{aligned}$$

$$= \Pr\left(\gamma_E = \max(\gamma_E^2, \gamma_E^3) \geq \gamma_{th} = 2^{\frac{2R}{(1-\alpha)}} - 1\right), \quad (13)$$

where  $\gamma_{th} = 2^{\frac{2R}{(1-\alpha)}} - 1$  and  $R$  [bits/s/Hz] is the expected rate. To guarantee both reliability and security, both metrics must be simultaneously minimized; this technique is a departure from other works in the literature which minimize the OP and IP separately. The bi-objective function is also explicitly expressed through rigorous mathematics rather than abstract form as in state-of-the-art techniques [8], [10]. To do this, we first derive the mathematical framework for the OP and IP using the following two theorems.

*Theorem 1:* Let us consider  $\mathcal{M}$  independent and non-identically distributed (i.n.i.d.) exponential random variables denoted by  $X_m$ ,  $m \in \{1, \dots, \mathcal{M}\}$ , with parameters  $\lambda_m$ . The cumulative distribution function (CDF) of the maximal RVs, i.e.,  $Y = \max_{m=1,2,\dots,\mathcal{M}}(X_m)$  is then computed as:

$$F_Y(y) = 1 + \sum \exp\left(-\sum_{t=1}^m \frac{y}{\lambda_{n_t}}\right), \quad (14)$$

$$\text{where } \sum = \sum_{m=1}^{\mathcal{M}} \frac{(-1)^m}{m!} \sum_{n_1=1}^{\mathcal{M}} \dots \sum_{n_m=1}^{\mathcal{M}}.$$

$n_1 \neq n_2 \dots \neq n_m$

*Proof:* The proof is given in Appendix I.  $\square$

*Theorem 2:* Let us consider  $\mathcal{M}$  i.n.i.d. non-central chi square RVs denoted by  $X_m$ ,  $m \in \{1, \dots, \mathcal{M}\}$ , with parameters  $K_m$  and  $a_m$ . The CDF of the maximal RVs, i.e.,  $Y = \max_{m=1,2,\dots,\mathcal{M}}(X_m)$  is then computed as:

$$F_Y(y) = 1 + \sum \frac{(K_{n_t})^{i_t} (a_{n_t})^{j_t}}{i_t! j_t!} y^{\sum_{t=1}^m j_t} \times \exp\left(-\sum_{t=1}^m (K_{n_t} + a_{n_t} y)\right), \quad (15)$$

$$\text{where } \sum = \sum_{m=1}^{\mathcal{M}} \frac{(-1)^m}{m!} \sum_{n_1=1}^{\mathcal{M}} \dots \sum_{n_m=1}^{\mathcal{M}} \sum_{i_1 \geq 0}^{i_1} \dots \sum_{i_m \geq 0}^{i_m} \sum_{j_m=0}^m \prod_{t=1}^m.$$

$n_1 \neq n_2 \dots \neq n_m$

*Proof:* The proof is given in Appendix II.  $\square$

Having obtained the CDF of the maximal of the exponential and non-central chi-square RVs, the OP and IP of the proposed network are computed in Sections 3 and 4 which are given below.

#### A. Outage Probability

The exact and approximate closed-form expression at the destination are given in Theorem 3.

*Theorem 3:* Let us denote  $\sum_{l_n, n \in (1, \dots, N)} =$

$$\sum_{n=1}^N \frac{(-1)^n}{n!} \sum_{l_1=1}^N \dots \sum_{l_n=1}^N;$$

$l_1 \neq l_2 \dots \neq l_n$

$$\sum_{i_m, n_m, m \in (1, \dots, M)} = \sum_{m=1}^M \frac{(-1)^m}{m!} \sum_{n_1=1}^M \dots \sum_{n_m=1}^M \sum_{i_1 \geq 0}^{i_1} \dots \sum_{i_m \geq 0}^{i_m} \sum_{j_m=0}^m \prod_{t=1}^m,$$

$n_1 \neq n_2 \dots \neq n_m$

whereby the OP under the exact framework is given by (16), as shown at the bottom of the page.

Here  $\mathcal{K}_v(\bullet)$  is the modified Bessel function of second kind with  $v$ -th order [30, Eq. 8.407.2].

*Proof:* The proof is given in Appendix III.  $\square$

Direct inspection (16) indicates that the integral cannot be computed in the closed-form expression, therefore it is difficult to gain any insight based on the rigorous mathematical framework. To overcome this problem, we propose an approximate framework where the e2e SINR at the destination in (13) is approximated as

$$\gamma_{D_c}^{\text{ap}} = \frac{\mu\Phi\Psi|h_{S_bR}|^2|h_{RD_c}|^2|h_{BR}|^2}{\mu\Psi|h_{RD_c}|^2|h_{BR}|^2 + \Phi|h_{S_bR}|^2} = \frac{\mu\Phi\Psi|h_{S_bR}|^2\gamma_{BRD_c}}{\mu\Psi\gamma_{BRD_c} + \Phi|h_{S_bR}|^2} \approx \Phi \min\left(|h_{S_bR}|^2, \frac{\gamma_{BRD_c}}{\tilde{\mu}}\right), \quad (17)$$

where  $\tilde{\mu} = \frac{\Phi}{\mu\Psi}$  and use the shorthand  $\gamma_{BRD_c} = |h_{BR}|^2|h_{RD_c}|^2$ . The closed-form expression of the OP under the approximation e2e SINR at D in (17) denoted by  $\widetilde{\text{OP}}$  is then computed from

$$\begin{aligned} \widetilde{\text{OP}} = 1 + 2 \sum_{l_n, n \in (1, \dots, N)}^{\bullet} \sum_{i_m, n_m, m \in (1, \dots, M)}^{\bullet\bullet} & \times \frac{(K_{n_t})^{i_t} (a_{n_t})^{j_t}}{i_t! j_t!} \left(\frac{\gamma_{\text{th}}}{\Phi}\right)^{\sum_{t=1}^m j_t} \sqrt{\sum_{t=1}^n \frac{\tilde{\mu}\gamma_{\text{th}}}{\Phi\lambda_{BR}\lambda_{RD_t}}} \\ & \times \exp\left(-\sum_{t=1}^m \left[K_{n_t} + \frac{a_{n_t}\gamma_{\text{th}}}{\Phi}\right]\right) \\ & \times \mathcal{K}_1\left(2\sqrt{\sum_{t=1}^n \frac{\tilde{\mu}\gamma_{\text{th}}}{\Phi\lambda_{BR}\lambda_{RD_t}}}\right), \end{aligned} \quad (18)$$

*Proof:* The proof is given in Appendix IV.  $\square$

*Remark 2:* Examining (18), the impact of  $\Phi$  and  $\Psi$  is straightforward as large-scale path-loss from  $PB$  to  $R$  and from  $R$  to  $D$ , whereas in (16), it is impossible to form any conclusions since these terms are inside the integration.

The next section addresses the security aspect of the proposed networks using the intercept probability metric.

### B. Intercept Probability

Whereas the OP addresses the reliability aspect of the considered networks, the IP focuses on security issues, i.e.,

the IP measures the probability that an eavesdropper listens in on legitimate information. As with the OP, both exact and approximate frameworks are obtained from Theorem 4.

*Theorem 4:* Let us define shorthand  $\Lambda_4(x) = 1 - \sum_{l \geq 0} \sum_{p=0}^l \frac{(K_m)^l (a_m)^p}{l! p!} \left(\frac{\gamma_{\text{th}}(\Psi x + 1)}{\Phi}\right)^p \times \exp\left(-K_m - a_m \frac{\gamma_{\text{th}}(\Psi x + 1)}{\Phi}\right)$ . Denoted by IP, the exact framework of the IP at the eavesdropper is computed from (19), as shown at the bottom of the next page.

*Proof:* The proof is given in Appendix V.  $\square$

Examining (19), the integral is too complicated to compute in closed-form since it involves a special function, a complicated argument in the exponential functions, and rationale functions, and hence, it is not feasible to gain any insight from these mathematical frameworks. We, therefore, propose a tight approximation of the closed-form expression for IP, as with the OP. We ignore particularly the impact of background noise and the term  $\Phi|h_{S_bR}|^2$  in the e2e SINR of  $E$  in (10) since it is not comparable to any other terms. Denoted by  $\widetilde{\text{IP}}$ , the approximate framework of the IP is then computed from (20), as shown at the bottom of the next page.

*Proof:* The proof is given in Appendix VI.  $\square$

Having obtained the closed-form expression of the OP and IP, we are interested in simultaneously minimizing the values of both metrics. The rationale behind simultaneous minimization is the following: for example, if we increase the TS ratio, the relay will harvest more energy and therefore the probability that the transmission from  $R$  to  $D$  is successful, which decreases the OP, is high. However, the higher the transmit power at  $R$  the higher the intercept probability. Regarding the UAV's position, if the UAV is near the eavesdropper, the OP and IP will undoubtedly approach 1 concurrently, unless  $E$  is very close to  $D$ . If the UAV is too far from  $R$ , however, the OP and IP will also quickly approach 1. As a result, an optimal UAV position and the TS ratio for simultaneously minimizing two metrics exist; joint minimization of both the OP and IP with respect to the UAV's position and TS ratio is therefore required.

## IV. JOINT OP AND IP MINIMIZATION: A MULTI-OBJECTIVE OPTIMIZATION APPROACH

The current study departs from other works that minimize the OP and IP separately by instead minimizing these metrics concurrently, according to the approximate closed-form

$$\begin{aligned} \text{OP} = 2 + 2 \sum_{l_n, n \in (1, \dots, N)}^{\bullet} & \sqrt{\left(\frac{\gamma_{\text{th}}}{\mu\Psi}\right) \sum_{t=1}^n \frac{1}{\lambda_{BR}\lambda_{RD_t}}} \mathcal{K}_1\left(2\sqrt{\left(\frac{\gamma_{\text{th}}}{\mu\Psi}\right) \sum_{t=1}^n \frac{1}{\lambda_{BR}\lambda_{RD_t}}}\right) \\ & - 2 \sum_{v_n, n \in (1, \dots, N)}^{\bullet} \sum_{i_m, n_m, m \in (1, \dots, M)}^{\bullet\bullet} \left(\sum_{w=1}^n \frac{1}{\lambda_{BR}\lambda_{RD_w}}\right) \frac{(K_{n_t})^{i_t} (a_{n_t})^{j_t}}{i_t! j_t!} \left(\frac{\gamma_{\text{th}}\mu\Psi}{\Phi}\right)^{\sum_{t=1}^m j_t} \\ & \times \int_{\frac{\gamma_{\text{th}}}{\mu\Psi}}^{+\infty} \left(\frac{x}{\mu\Psi x - \gamma_{\text{th}}}\right)^{\sum_{t=1}^m j_t} \exp\left[-\sum_{t=1}^m \left(K_{n_t} + a_{n_t} \frac{\gamma_{\text{th}}\mu\Psi x}{\Phi(\mu\Psi x - \gamma_{\text{th}})}\right)\right] \mathcal{K}_0\left(2\sqrt{x \sum_{w=1}^n \frac{1}{\lambda_{BR}\lambda_{RD_w}}}\right) dx \end{aligned} \quad (16)$$

expressions described in Section III, and thus optimizing the position of the UAV and time-switching ratio simultaneously. The optimization problem is formulated as

$$\begin{aligned} & \underset{\alpha, x_S, y_S, z_S}{\text{minimize}} && f_0(\alpha, x_S, y_S, z_S), \\ & \text{subject to} && \alpha_{\min} \leq \alpha \leq \alpha_{\max}, \\ & && x_{\min} \leq x_S \leq x_{\max}, \\ & && y_{\min} \leq y_S \leq y_{\max}, \\ & && z_{\min} \leq z_S \leq z_{\max}, \end{aligned} \quad (21)$$

where  $f_0(\alpha, x_S, y_S, z_S) = [\widetilde{\text{OP}}(\alpha, x_S, y_S, z_S), \widetilde{\text{IP}}(\alpha, x_S, y_S, z_S)]^T \in \mathbb{R}_+^2$  is the objective function for the two different measurement metrics. The OP and IP expressions are defined in (18) and (20), respectively,  $\alpha_{\min}, \alpha_{\max}$  are the upper and lower bounds of the time-switching factor  $\alpha$ , and  $(x_{\min}, y_{\min}, z_{\min})$  and  $(x_{\max}, y_{\max}, z_{\max})$  denote the limiting coordinates of  $(x_S, y_S, z_S)$ . We stress that the problem (21) contains novel contributions because it handles the two different practical objective functions simultaneously and the prior information is the statistical channel information only, which is stable over multiple coherence blocks. Inspecting (21), it is clear that the problem is non-convex since the utility function is non-convex. Nevertheless, the OP and IP are continuous functions, and the feasible domain is a convex set. The global optimum, therefore, exists by virtue of the Weierstrass' theorem [35].

To solve this bi-objective optimization problem, we adopt a well-known multi-objective algorithm, known as the non-dominated sorting genetic algorithm version II [31]. The

algorithm provides a local optimum solution. We selected the NSGA-II to solve the joint OP and IP minimization problem because the literature shows that it is one of the most effective and stable evolutionary multi-objective algorithms available [32]. The NSGA-II framework is based on two main principles, one being fast non-dominated sorting for ranking the individual multiple objective values, the other being crowding distance for diversifying the solutions of the output Pareto front. These principles are briefly described in the subsections below.

#### A. Non-Dominated Sorting

We first recall the definition of Pareto domination. Like other evolutionary algorithms, NSGA-II maintains a population  $P$  of  $N$  individuals throughout the search process, where individual  $i$  is associated with a 4-dimensional vector  $I_i = (\alpha_i, x_{iR}, y_{iR}, z_{iR})$  that represents a solution to the problem (21). A solution  $I_i$  is said to dominate or is better than a solution  $I_j$  if one of the following conditions holds:

$$\widetilde{\text{OP}}(I_i) \leq \widetilde{\text{OP}}(I_j) \quad \text{and} \quad \widetilde{\text{IP}}(I_i) < \widetilde{\text{IP}}(I_j), \quad (22)$$

which improves the IP, at least, or

$$\widetilde{\text{OP}}(I_i) < \widetilde{\text{OP}}(I_j) \quad \text{and} \quad \widetilde{\text{IP}}(I_i) \leq \widetilde{\text{IP}}(I_j), \quad (23)$$

which improves the OP, at least. With this in mind, the first-ranked Pareto front is defined as the set of all individuals not dominated by any others, the second-ranked front as the set of individuals only dominated by those in the first front, and so on.

$$\begin{aligned} \text{IP} &= 1 - \int_0^{+\infty} (1 - \Lambda_4(x) \Lambda_5(x)) f_{|h_{\text{BE}}|^2}(x) dx \\ \Lambda_5(x) &= \left[ 1 - 2 \sqrt{\frac{\gamma_{\text{th}}(\Psi x + 1)}{\mu \Psi \lambda_{\text{BR}} \lambda_{\text{RE}}}} K_1 \left( 2 \sqrt{\frac{\gamma_{\text{th}}(\Psi x + 1)}{\mu \Psi \lambda_{\text{BR}} \lambda_{\text{RE}}}} \right) \right] \\ &\quad + \frac{2}{\lambda_{\text{BR}} \lambda_{\text{RE}}} \int_{\frac{\gamma_{\text{th}}(\Psi x + 1)}{\mu \Psi}}^{\infty} \mathcal{K}_0 \left( 2 \sqrt{\frac{y}{\lambda_{\text{BR}} \lambda_{\text{RE}}}} \right) \\ &\quad \times \left( 1 - \sum_{\substack{i_m, n_m, \\ m \in (1, \dots, M)}}^{\bullet\bullet} \times \frac{(K_{n_t})^{i_t} (a_{n_t})^{j_t}}{i_t! j_t!} \left( \frac{\mu \Psi \gamma_{\text{th}} y}{\Phi (\mu \Psi y - \gamma_{\text{th}} (\Psi x + 1))} \right)^{\sum_{t=1}^m j_t} \right. \\ &\quad \left. \times \exp \left( - \sum_{t=1}^m \left( K_{n_t} + \frac{a_{n_t} \mu \Psi \gamma_{\text{th}} y}{\Phi (\mu \Psi y - \gamma_{\text{th}} (\Psi x + 1))} \right) \right) \right) dy \end{aligned} \quad (19)$$

$$\begin{aligned} \widetilde{\text{IP}} &= 1 - \left[ 1 - \sum_{l \geq 0} \sum_{p=0}^l \frac{\exp(-K_m) (K_m)^l (a_m)^p}{\lambda_{\text{BE}} l!} \left( \frac{\gamma_{\text{th}} \Psi}{\Phi} \right)^p \left( \frac{a_m \gamma_{\text{th}} \Psi}{\Phi} + \frac{1}{\lambda_{\text{BE}}} \right)^{-1-p} \right] \left[ 1 - \exp \left( \frac{\lambda_{\text{BE}}}{2 \lambda_{\text{BR}} \lambda_{\text{RE}}} \frac{\gamma_{\text{th}}}{\mu} \right) \right. \\ &\quad \left. \times W_{-1, \frac{1}{2}} \left( \frac{\lambda_{\text{BE}}}{\lambda_{\text{BR}} \lambda_{\text{RE}}} \frac{\gamma_{\text{th}}}{\mu} \right) \left( \sum_{l \geq 0} \sum_{p=0}^l \frac{(K_m)^l (a_m)^p}{l! p!} \left( \frac{\gamma_{\text{th}}}{\Phi} \right)^p \exp \left( -K_m - a_m \frac{\gamma_{\text{th}}}{\Phi} \right) \right) \right] \end{aligned} \quad (20)$$



**Algorithm 1** NSGA-II Based OP/IP Minimization

---

**Input:**  $N, M, \Phi, \Psi, \mu, K, a, \gamma_{th}, \lambda_{BE}, \lambda_{BR}, \lambda_{RE}, \lambda_{RD}, \tilde{\mu}$

- 1: Randomly generate and evaluate  $N$  individuals of the initial population  $P_0$ .
- 2: Set the number of generations  $g \leftarrow 0$ .
- 3: **while**  $g < g_{max}$  **do**
- 4:   Execute crossover and mutation on  $P_g$  to generate offspring population  $O_g$ .
- 5:    $Q_g \leftarrow P_g \cup O_g$ .
- 6:   Divide  $Q_g$  into Pareto fronts  $F_1, F_2, \dots$  using the non-dominated sorting procedure.
- 7:   Find  $i^*$  such that  $\sum_{i=1}^{i^*-1} |F_i| \leq N < \sum_{i=1}^{i^*} |F_i|$ .
- 8:    $P_{g+1} \leftarrow \bigcup_{i=1}^{i^*-1} F_i$ .
- 9:    $C \leftarrow$  Select  $N - \sum_{i=1}^{i^*-1} |F_i|$  individuals with highest crowding distance from  $F_{i^*}$ .
- 10:    $P_{g+1} \leftarrow P_{g+1} \cup C$ .
- 11:    $g \leftarrow g + 1$ .
- 12: **end while**
- 13: **return** The first-ranked Pareto front.

---

In each generation of NSGA-II, the ranks of individuals, which are equal to the rank of their corresponding Pareto front, are determined by sorting the population based on non-domination. Each individual is first compared to all others to identify non-dominated members. These non-dominated individuals are then assigned to the first-ranked Pareto front and obtain rank one. Next, all the individuals of the first front are temporarily removed from the population. The comparison is performed on the remaining population to extract solutions of the second-ranked front. The procedure continues until the ranks of all individuals have been determined.

**B. Crowding Distance**

It is desired that the algorithm maintains diversity in the solutions for the Pareto front. In the case of the bi-objective such as in the current study, the crowding distance of a solution  $i$  is defined as the average distance between two nearest solutions on both its sides. This quantity indicates the density of solutions in the space of objective values and is used as a metric in the selection step along with the individuals' ranks. Specifically, from the pool of individuals with different ranks, those with lower ranks are selected. If the candidates have the same rank, those with higher crowding distance are preferred.

**C. Algorithm Structure**

Using the above descriptions, the procedure for jointly minimizing the IP and OP is presented in Algorithm 1. The algorithm begins by generating  $N$  individuals in the feasible domain using the uniform distribution. In each generation, two genetic operators, called Binary simulation crossover [33] and Polynomial mutation [34], are performed to generate the offspring. Elitist selection based on the ranks of Pareto fronts and crowding distance is then invoked to select individuals of the next generation. Overall, the algorithm requires  $O(\tilde{g}MN^2)$  computations, where  $\tilde{g}$  is the number of iterations required to

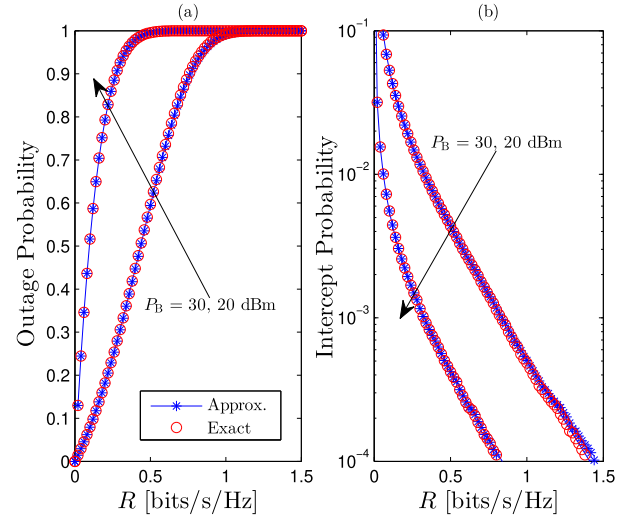


Fig. 2. Outage probability (a) and Intercept probability (b) versus  $R$  [bits/s/Hz], for various values of  $P_B$ . Solid lines are derived from (16), (18), (19) and (20), and markers indicate Monte-Carlo simulations.

reach the convergence,  $M$  is the number of objectives, and  $N$  is the population size. The computational complexity order indicates that Algorithm 1 provides the solution in polynomial time for finite-dimensional networks.

**V. NUMERICAL RESULTS**

This section provides numerical results that verify the accuracy of the derived mathematical framework and substantiate the significance of optimal UAV positioning and the time-switching factor. Without loss of generality, the following parameters are applied throughout this section:  $\delta = 2.2$ ,  $\mathcal{M} = 4$ ,  $\alpha = 0.65$ ,  $\eta = 0.55$ ,  $R = 0.25$ ,  $K = 5$ ,  $\mathcal{N} = 4$ ,  $Bw = 5$  MHz,  $NF = 6$  dB,  $f_c = 1.8$  GHz,  $P_B = 30$  dBm,  $P_S = 10$  dBm.

Fig. 2 illustrates the performance of the OP (a) and IP (b) with respect to the expected rate,  $R$  [bits/s/Hz]. We observe that the simulation results are absolutely aligned with our derived mathematical framework. Additionally, the gaps between the exact (denoted by “Exact”) and approximated frameworks (denoted by “Approx.”) are indistinguishable. It is clear that increasing the expected rate will increase the OP and decrease the IP. This is directly explained from the definitions of the OP and IP. We observe that the greater the transmit power of the power beacon the smaller the values of both OP and IP. Particularly, when  $P_B$  increases to 10 dBm, the OP improves approximately two-fold at  $R = 0.5$  [bits/s/Hz]; the improvement in IP is even more impressive, almost tenfold, when  $R = 0.5$  [bits/s/Hz]. The detailed effects of  $P_B$  on the performance of OP and IP are plotted in Fig. 3.

Fig. 3 charts the behaviors of OP (a) and IP (b) versus the transmit of the power beacon  $P_B$ , with different values of  $R$  and  $P_S$ . We observe again that the derived mathematical frameworks consistently agree with the Monte-Carlo simulation results. Moreover, the differences between the exact and approximation frameworks are minor, even with low transmit power when the systems probably operate in a noise-limited

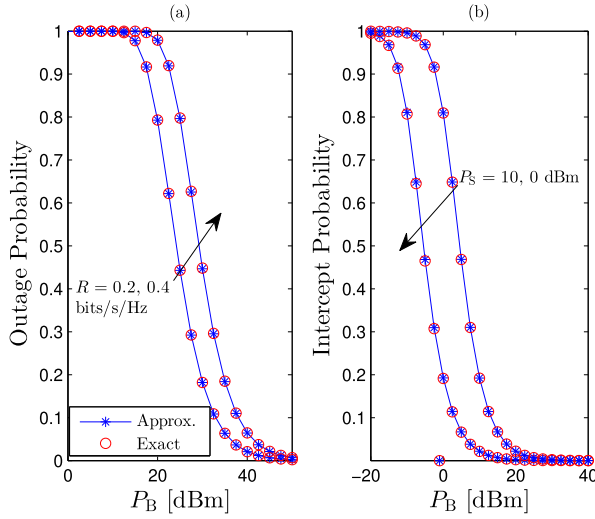


Fig. 3. Outage probability (a) and Intercept probability (b) versus  $P_B$  [dBm] for various values of  $P_S$  and  $R$ . Solid lines are derived from (16), (18), (19) and (20), and markers indicate Monte-Carlo simulations.

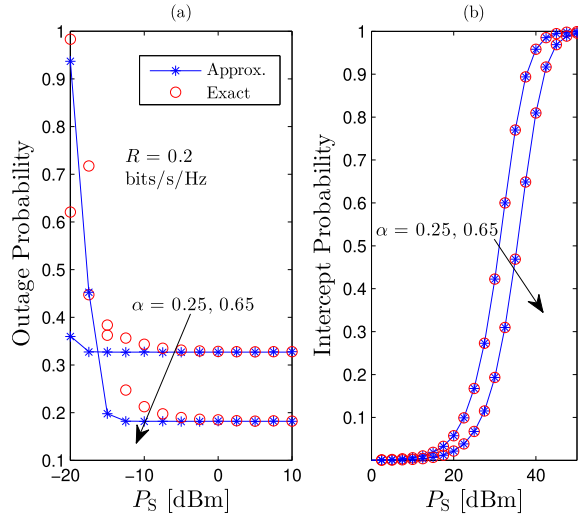


Fig. 4. Outage probability (a) and Intercept probability (b) versus  $P_S$  [dBm] for various values of  $\alpha$  and  $R$ . Solid lines are derived from (16), (18), (19) and (20), and markers indicate Monte-Carlo simulations.

regime. Fig. 3(a) confirms the findings in Fig. 2(a), where increasing the expected rate is harmful to the OP. As for the impact of the  $P_S$  on the performance of IP, we observe that increasing the  $P_S$  degrades the security of the system to the extent that the eavesdropper has a greater probability of listening in on secure information transmitted from the UAV to  $D$ . Figure 4 provides detail of the impact of the UAV's transmit power.

Fig. 4 plots the performance of OP (a) and IP (b) in relation to the UAV's transmit power  $P_S$ . It is interesting to note that increasing  $P_B$  does not consistently improve the performance of the OP. The rationale behind this phenomenon is that increasing the  $P_S$  simply facilitates the performance of the first hop from the UAV to  $R$ , thus the performance of the entire system is constrained by the second hop from  $R$  to  $D$ . We also observe a relatively large gap between the exact and approximation framework when the transmit power of  $P_S$

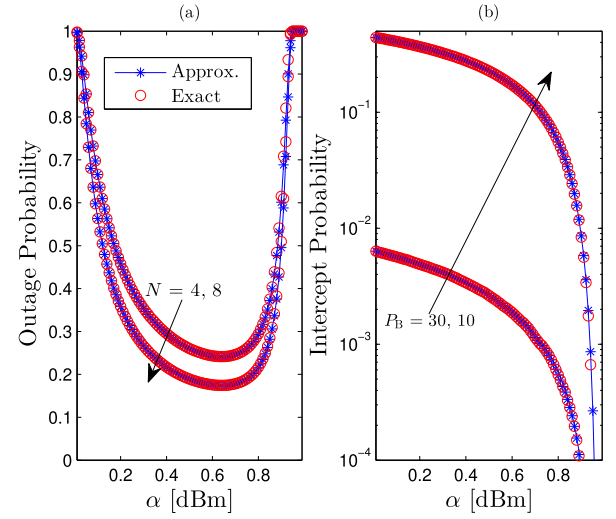


Fig. 5. Outage probability (a) and Intercept probability (b) versus  $\alpha$  for various values of  $P_B$  and  $N$ . Solid lines are derived from (16), (18), (19) and (20), and markers indicate Monte-Carlo simulations.

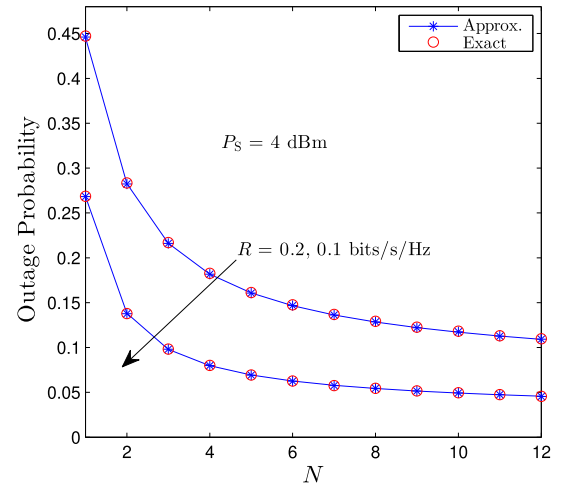


Fig. 6. Outage probability (a) and Intercept probability (b) versus  $N$  for various values of  $R$ . Solid lines are derived from (16) and (18), and markers indicate Monte-Carlo simulation.

is extremely small. However, no gaps are evident between the exact and approximation frameworks, even with a small transmit power. The trend of IP versus  $P_S$  also contrasts with the OP, which monotonically increases  $P_S$ . This means that the higher the  $P_S$  the worse the system security. The main reason for obtaining a higher IP is that a greater transmit power produces a better signal at the eavesdropper, hence, the higher the intercept probability. Fig. 4 reveals that increasing  $\alpha$  monotonically improves the OP and IP. But does this trend always hold? Fig. 5 provides an answer.

The impact of  $\alpha$  on the performance of OP (a) and IP (b) is given in Fig. 5. We see that scaling up the time-switching ratio substantially improves the performance of IP. It is not the case, however, for the OP: increasing  $\alpha$ , the OP first decreases after approaching its peak, then increases and approaches 1 when  $\alpha \rightarrow 1$ .

Fig. 6 depicts the impact of the number of antennae at destinations. It is clear that boosting  $N$  decreases the OP, yet

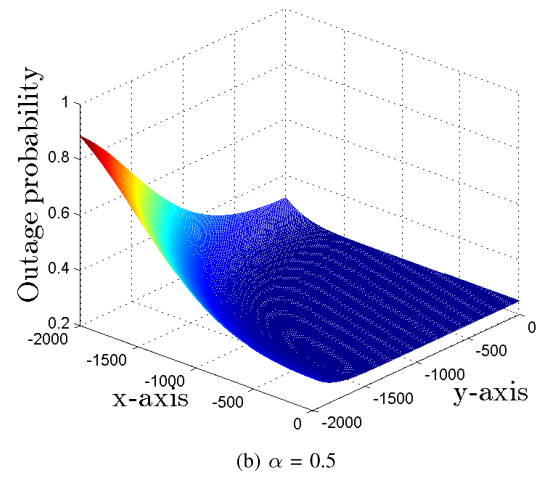
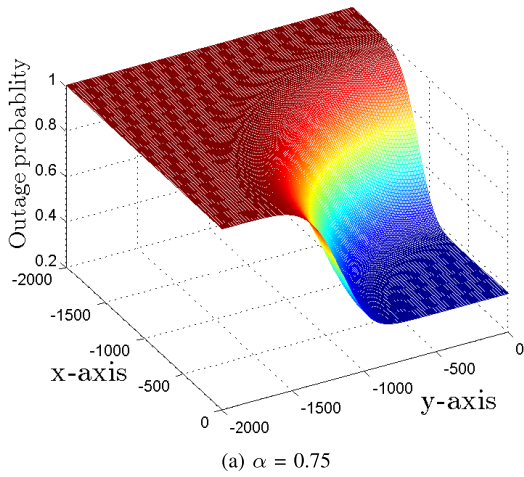


Fig. 7. Outage probability versus  $x$  &  $y$ . Lines are derived from (16) and (18).

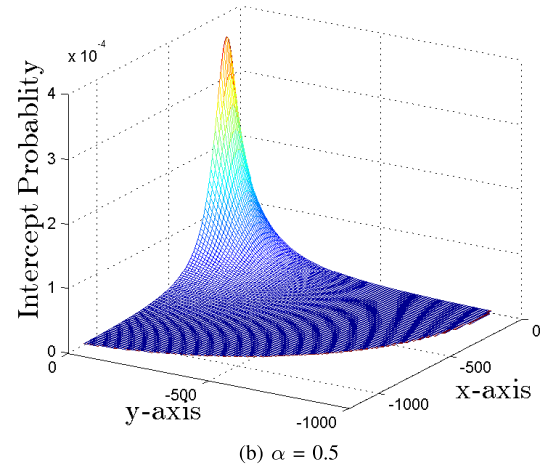
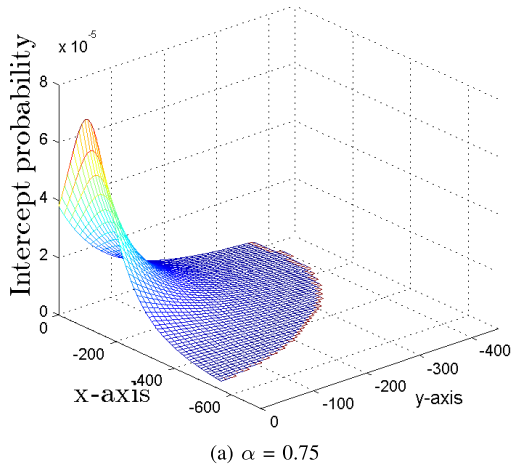


Fig. 8. Intercept probability vs.  $x$  &  $y$ -axis of UAV. Lines are derived from (19) and (20).

the increasing pace is divergent when  $\mathcal{N}$  is small and  $\mathcal{N}$  is large. When  $\mathcal{N}$  increases 1 to 6, the OP drops from 0.45 to 0.15; when  $\mathcal{N}$  increases from 7 to 12, the decrease in OP is less significant from 0.15 to 0.11.

Fig. 7 illustrates the effect of the  $x$ -axis and  $y$ -axis positions of the UAV, with different values for the time-switching ratio  $\alpha$ . We observe that different values of  $\alpha$  produce different shapes of the OP. For  $\alpha = 0.75$ , the OP approaches 1 when either  $x$  or  $y$  approaches  $-2000$ . However, for  $\alpha = 0.25$ , the OP is not necessarily equal to 1 when either  $x$  or  $y$ -axis approaches 0. It is therefore important to simultaneously optimize the position of the UAV and  $\alpha$ . Fig. 8 indicates the performance of IP with respect to the horizontal and vertical position of the UAV for various values of  $\alpha$ . Unlike the OP, we observe the same shape for two values of  $\alpha$ . Besides, when the UAV is close to the relay, the OP is worse. It is noted that the results illustrated in Figs. 7 and 8 are not based on the optimal coordinates of the source node. The main aim of these figures is to show the significance of optimizing the source coordinates and TS ratio since the OP and IP exhibit contrary behaviors regarding the  $S$  position and time-switching ratio. As a consequence, there exists an optimal set of the

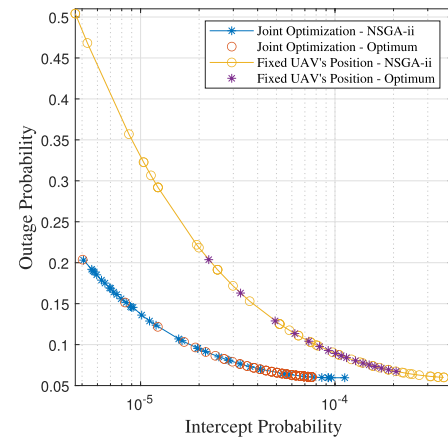


Fig. 9. The Pareto frontier obtained by solving the problem (21) and its special case by fixing the UAV's position. The proposed solutions are also compared with the global optimum obtained by an exhaustive search.

source position and time-switching ratio that simultaneously minimizing the OP and IP. The performance of the OP and IP with respect to the optimal coordinates of the source node and TS ratio is given in Fig. 9.

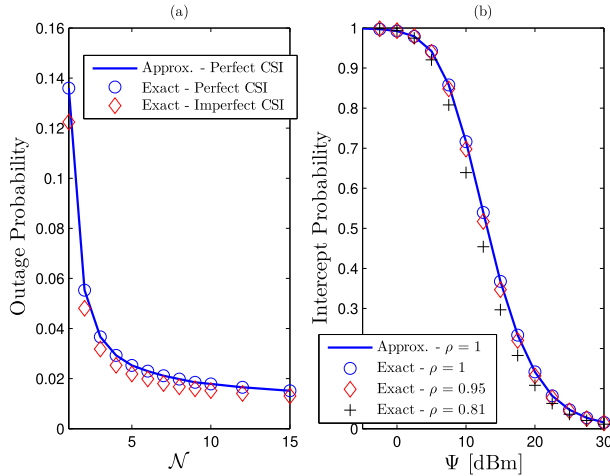


Fig. 10. OP vs.  $\mathcal{N}$  (a) and IP vs.  $\Psi$  (b) under the impact of imperfect channel state information. Solid lines are derived from (16), (18), (19) and (20) while markers are Monte-Carlo simulation.

Fig. 9 indicates the Pareto frontier for a system with limiting values for the time-switching factor  $\alpha_{\min} = 0.1$  and  $\alpha_{\max} = 0.9$ . The limiting coordinates are  $(x_{\min}, y_{\min}, z_{\min}) = (-1000, -1000, 1)$  [m] and  $(x_{\max}, y_{\max}, z_{\max}) = (0, 0, 200)$ . We compare the optimal solution obtained by jointly optimizing the time-switching factor  $\alpha$  and the UAV's coordinate and only considering  $\alpha$  as the optimization variable. In solving the problem (21), the results indicate superior improvements in boosting communications reliability. For the given IP of  $4.15 \times 10^{-5}$ , the OP is about 0.07 because both  $\alpha$  and  $(x, y, z)$  have been optimized. If the system only optimizes the UAV's coordinates, the OP is about 0.149, which is  $2 \times$  worse than the best solution. Not shown in Fig. 9, however, is that a fixed value of the time-switching factor  $\alpha$  usually yields a bad Pareto frontier. To thoroughly examine the algorithm's performance, we have compared the Pareto fronts obtained by the proposed algorithm and the optimal Pareto fronts, which are all depicted in Fig. 9. The optimal Pareto fronts are obtained by performing a grid search procedure that exhaustively generates and evaluates the solutions from a grid of parameter values. To obtain the (mostly) optimal Pareto fronts, we first separately calculate the IP and OP values for all solutions from the grid. The Pareto front is then constructed by combining the calculated IP and OP values with a weight that varies within the unit interval. Let  $K_{\alpha}$ ,  $K_x$ ,  $K_y$  and  $K_z$  be the number of sampled values for the parameters  $\alpha$ ,  $x_s$ ,  $y_s$  and  $z_s$ , respectively. The grid search method then requires evaluating  $K_{\alpha} K_x K_y K_z$  solutions. In our experiments, we set  $K_{\alpha} = 100$ ,  $K_x = 200$ ,  $K_y = 200$  and  $K_z = 100$ , resulting in a total of  $4 \times 10^8$  solutions needed to be evaluated. This is significantly more expensive compared to  $g_{\max} N$  solutions required in the proposed algorithms (which amounted to  $50 \times 50 = 2500$  solutions in our setup). Fig. 9 shows that the proposed algorithm performed remarkably well as the Pareto fronts obtained by the algorithm overlap with the optimal fronts.

Fig. 10 depicts the performance of the OP versus the number of received antennae,  $\mathcal{N}$  (a), and the IP with respect to the transmit power of the power beacon  $\Psi$  (b), under the impact

of the imperfect channel estimation. Particularly, we employ the imperfect channel state information as in [36] where the channel coefficient of an arbitrary link from transmitter  $p \in \{S, R\}$  to receiver  $q \in \{R, E, D\}$  is formulated as  $\tilde{h}_{p,q} = \rho h_{p,q} + \sqrt{1 - \rho^2} v_{p,q}$ , where  $v_{p,q}$  is a complex Gaussian RV with zero mean and the same variance as  $h_{p,q}$ . Here  $\tilde{h}_{p,q}$  is the imperfect version of  $h_{p,q}$  and  $\rho \in [0, 1]$  is the correlation coefficient. The CSI is called perfect providing that  $\rho = 1$ . We observe that by increasing the number of received antennae, the OP keeps decreasing thus improving the reliability. Although imperfect CSI has a harmful effect on the OP, the gap between the perfect and imperfect CSI is minor even with  $\mathcal{N} = 1$  and steadily declines when  $\mathcal{N}$  keeps going up. The influences of the imperfect CSI on the IP performance, on the contrary, is useful. We observe on Fig. 10(b) that the smaller  $\rho$  the smaller the value of IP. It signifies that the higher the security of the system. This figure also confirms the benefits of the friendly jamming technique that by increasing the AN at the eavesdropper one can dramatically decrease the IP.

## VI. CONCLUSION

The current study investigated the trade-off between security and reliability in UAV terrestrial communications aided by an EH-enabled relay. The exact and approximate tight closed-form expressions of the OP and IP were derived. To study the effects of the UAV's position and TS ratio, a bi-objective function was formulated. The OP and IP were minimized with respect to the UAV's position and TS ratio. It should be noted that the study considered a true 3-D position for the UAV rather than a fixed altitude, unlike other works in the literature. A non-dominated sorting genetic algorithm version II was applied to obtain a sub-optimal solution. The results of the study indicate that by appropriately positioning the UAV and specifying the TS ratio, security and reliability targets can be achieved concurrently.

## APPENDIX I PROOF OF EQUATION (14)

Here, we derive the CDF of  $\mathcal{M}$  independent and non-identically distributed exponential random variables. Thus,

$$\begin{aligned}
 F_Y(y) &= \Pr \left( Y = \max_{m=1,2,\dots,M} (X_m) < y \right) = \prod_{m=1}^M \left( 1 - \exp \left( -\frac{y}{\lambda_m} \right) \right) \\
 &\stackrel{(a)}{=} 1 + \sum_{m=1}^M \frac{(-1)^m}{m!} \underbrace{\sum_{n_1=1}^M \dots \sum_{n_m=1}^M}_{n_1 \neq n_2 \dots \neq n_m} \left[ \prod_{t=1}^m \exp \left( -\frac{y}{\lambda_{n_t}} \right) \right] \\
 &= 1 + \sum \dot{\exp} \left( -\sum_{t=1}^m \frac{y}{\lambda_{n_t}} \right), \tag{24}
 \end{aligned}$$



where (a) is obtained by employing the following identity  $\prod_{m=1}^M (1 - x_m) = 1 + \sum_{m=1}^M \frac{(-1)^m}{m!} \sum_{n_1=1}^M \dots \sum_{n_m=1}^M \left[ \prod_{t=1}^m x_{n_t} \right]$  and

$$\text{denote } \dot{\sum} = \sum_{m=1}^M \frac{(-1)^m}{m!} \sum_{n_1=1}^M \dots \sum_{n_m=1}^M \cdot$$

This ends the proof.

## APPENDIX II PROOF OF EQUATION (15)

Here, we derive the CDF of  $\mathcal{M}$  i.i.d. non-central chi-square RVs. Thus,

$$\begin{aligned} F_Y(y) &= \Pr \left( Y = \max_{m=1,2,\dots,M} (X_m) < y \right) \\ &\stackrel{(a)}{=} \prod_{m=1}^M \left( 1 - \exp(-K_m - a_m y) \sum_{l \geq 0} \sum_{p=0}^l \frac{(K_m)^l (a_m)^p}{l! p!} y^p \right) \\ &\stackrel{(b)}{=} 1 + \sum_{m=1}^M \frac{(-1)^m}{m!} \sum_{n_1=1}^M \dots \sum_{n_m=1}^M \sum_{i_1 \geq 0} \sum_{j_1=0}^{i_1} \dots \sum_{i_m=0}^{i_m} \sum_{j_m=0}^{i_m} \\ &\quad \times \left[ \prod_{t=1}^m \frac{(K_{n_t})^{i_t} (a_{n_t})^{j_t}}{i_t! j_t!} y^{t=1}^{j_t} \exp \left( - \sum_{t=1}^m (K_{n_t} + a_{n_t} y) \right) \right] \\ &= 1 + \dot{\sum} \frac{(K_{n_t})^{i_t} (a_{n_t})^{j_t}}{i_t! j_t!} y^{t=1}^{j_t} \exp \left( - \sum_{t=1}^m (K_{n_t} + a_{n_t} y) \right), \end{aligned} \quad (25)$$

where (a) is attained by substituting the CDF of the Rician distribution into (1), (b) is achieved by using the same identity in (24), and

$$\dot{\sum} = \sum_{m=1}^M \frac{(-1)^m}{m!} \sum_{n_1=1}^M \dots \sum_{n_m=1}^M \sum_{i_1 \geq 0} \sum_{j_1=0}^{i_1} \dots \sum_{i_m \geq 0} \sum_{j_m=0}^{i_m} \prod_{t=1}^m \cdot$$

This ends the proof.

## APPENDIX III PROOF OF EQ. (16)

Here, we derive the exact mathematical framework of the OP. Let us first rewrite the OP as

$$\begin{aligned} \text{OP} &= \Pr \left( \frac{\mu \Phi \Psi \gamma_{\text{SBR}} \gamma_{\text{BRD}_c}}{\mu \Psi \gamma_{\text{BRD}_c} + \Phi \gamma_{\text{SBR}}} < \gamma_{\text{th}} \right) \\ &= \Pr \left( \Phi \gamma_{\text{SBR}} (\mu \Psi \gamma_{\text{BRD}_c} - \gamma_{\text{th}}) < \gamma_{\text{th}} \mu \Psi \gamma_{\text{BRD}_c} \right) \\ &= \int_0^{\frac{\gamma_{\text{th}}}{\mu \Psi}} f_{\gamma_{\text{BRD}_c}}(x) dx + \int_{\frac{\gamma_{\text{th}}}{\mu \Psi}}^{\infty} F_{\gamma_{\text{SBR}}} \left( \frac{\gamma_{\text{th}} \mu \Psi x}{\Phi (\mu \Psi x - \gamma_{\text{th}})} \right) \\ &\quad \times f_{\gamma_{\text{BRD}_c}}(x) dx = \Lambda_1 + \Lambda_2. \end{aligned} \quad (26)$$

We then yield the following shorthand:  $\gamma_{\text{SBR}} = |h_{\text{SBR}}|^2$ ,  $\gamma_{\text{BRD}_c} = |h_{\text{BR}}|^2 |h_{\text{RD}_c}|^2$ . To compute  $\Lambda_1$  in

(26), we require the CDF of  $\gamma_{\text{BRD}_c}$ , which is computed as follows:

$$\begin{aligned} F_{\gamma_{\text{BRD}_c}}(x) &= \Pr(\gamma_{\text{BRD}_c} = \gamma_{\text{BR}} \gamma_{\text{RD}_c} < x) \\ &= \Pr \left( \gamma_{\text{RD}_c} < \frac{x}{\gamma_{\text{BR}}} \right) = \int_0^{+\infty} F_{\gamma_{\text{RD}_c}} \left( \frac{x}{y} \right) f_{\gamma_{\text{BR}}}(y) dy \stackrel{(a)}{=} 1 \\ &\quad + \dot{\sum}_{l_n, n \in (1, \dots, N)} \frac{1}{\lambda_{\text{BR}}} \int_{y=0}^{\infty} \exp \left( -\frac{y}{\lambda_{\text{BR}}} - \frac{x}{y} \sum_{t=1}^n \frac{1}{\lambda_{\text{RD}_t}} \right) dy \stackrel{(b)}{=} 1 \\ &\quad + 2 \dot{\sum}_{l_n, n \in (1, \dots, N)} \sqrt{x \sum_{t=1}^n \frac{1}{\lambda_{\text{BR}} \lambda_{\text{RD}_t}}} \mathcal{K}_1 \left( 2 \sqrt{x \sum_{t=1}^n \frac{1}{\lambda_{\text{BR}} \lambda_{\text{RD}_t}}} \right), \end{aligned} \quad (27)$$

where (a) is held by substituting the CDF of  $\gamma_{\text{RD}_c}$  from Theorem 1, and (b) is achieved with the assistance of [30, Eq. 3.471.9].  $\mathcal{K}_v(\bullet)$  is the modified Bessel function of the second kind with  $v$ -th order and  $\dot{\sum}_{l_n, n \in (1, \dots, N)} =$

$\sum_{n=1}^N \frac{(-1)^n}{n!} \sum_{l_1=1}^N \dots \sum_{l_n=1}^N$ . From (27), we immediately obtain  $\Lambda_1$ ,

given as

$$\begin{aligned} \Lambda_1 &= F_{\gamma_{\text{BRD}_c}} \left( \frac{\gamma_{\text{th}}}{\mu \Psi} \right) = 1 + 2 \dot{\sum}_{l_n, n \in (1, \dots, N)} \sqrt{\left( \frac{\gamma_{\text{th}}}{\mu \Psi} \right)} \\ &\quad \times \sqrt{\sum_{t=1}^n \frac{1}{\lambda_{\text{BR}} \lambda_{\text{RD}_t}}} \mathcal{K}_1 \left( 2 \sqrt{\left( \frac{\gamma_{\text{th}}}{\mu \Psi} \right) \sum_{t=1}^n \frac{1}{\lambda_{\text{BR}} \lambda_{\text{RD}_t}}} \right). \end{aligned} \quad (28)$$

Next, we derive the probability density function (PDF) of  $\gamma_{\text{BRD}_c}$ , where

$$\begin{aligned} f_{\gamma_{\text{BRD}_c}}(x) &= \frac{\partial F_{\gamma_{\text{BRD}_c}}(x)}{\partial x} = -2 \dot{\sum}_{l_n, n \in (1, \dots, N)} \\ &\quad \times \left( \sum_{t=1}^n \frac{1}{\lambda_{\text{BR}} \lambda_{\text{RD}_t}} \right) \mathcal{K}_0 \left( 2 \sqrt{x \sum_{t=1}^n \frac{1}{\lambda_{\text{BR}} \lambda_{\text{RD}_t}}} \right). \end{aligned} \quad (29)$$

Having obtained the CDF and PDF of  $\gamma_{\text{BRD}_c}$ , we compute  $\Lambda_2$  in (26) as follows:

$$\begin{aligned} \Lambda_2 &= 1 - 2 \dot{\sum}_{v_n, n \in (1, \dots, N)} \dot{\sum}_{i_m, n_m, m \in (1, \dots, M)} \left( \sum_{w=1}^n \frac{1}{\lambda_{\text{BR}} \lambda_{\text{RD}_w}} \right) \\ &\quad \times \frac{(K_{n_t})^{i_t} (a_{n_t})^{j_t}}{i_t! j_t!} \left( \frac{\gamma_{\text{th}} \mu \Psi}{\Phi} \right)^{\sum_{t=1}^m j_t} \int_{\frac{\gamma_{\text{th}}}{\mu \Psi}}^{+\infty} \left( \frac{x}{\mu \Psi x - \gamma_{\text{th}}} \right)^{\sum_{t=1}^m j_t} \\ &\quad \times \exp \left( - \sum_{t=1}^m \left( K_{n_t} + \frac{a_{n_t} \gamma_{\text{th}} \mu \Psi x}{\Phi (\mu \Psi x - \gamma_{\text{th}})} \right) \right) \end{aligned}$$

$$\times \mathcal{K}_0 \left( 2 \sqrt{x \sum_{w=1}^n \frac{1}{\lambda_{BR} \lambda_{RD_w}}} \right) dx. \quad (30)$$

Here, (30) is held by substituting the CDF of  $\gamma_{S_bR}$  into (29) and (26) and using the following short-hand  $\sum_{\substack{i_m, n_m, \\ m \in (1, \dots, M)}} =$

$$\sum_{m=1}^M \frac{(-1)^m}{m!} \sum_{n_1=1}^M \dots \sum_{n_m=1}^M \sum_{i_1 \geq 0}^{i_1} \dots \sum_{i_m \geq 0}^{i_m} \prod_{t=1}^m \dots. \text{ Directly}$$

inspecting (30), we find that integration is too complicated and consists of the special function, i.e., a modified Bessel function of the second kind, and the non-zero lower limits. We therefore cannot compute it in closed-form expression. However, it is feasible to compute (30) using numerical computations in popular computation software such as MATLAB or Mathematica. Finally, by substituting  $\Lambda_1$  and  $\Lambda_2$  into (26), we obtain (16). This ends the proof.

#### APPENDIX IV PROOF OF EQUATION (18)

Here, we derive the approximate mathematical framework of the OP. Let us begin with the approximate e2e SINR at D:

$$\gamma_{D_c}^{\text{ap}} = \Phi \min \left( |h_{S_bR}|^2, \frac{\gamma_{\text{BRD}_c}}{\tilde{\mu}} \right). \quad (31)$$

The OP under the approximate framework is then computed from (32):

$$\begin{aligned} \widetilde{\text{OP}} &= \Pr(\gamma_{D_c}^{\text{ap}} < \gamma_{\text{th}}) = \Pr\left(\Phi \min \left( |h_{S_bR}|^2, \frac{\gamma_{\text{BRD}_c}}{\tilde{\mu}} \right) < \gamma_{\text{th}}\right) \\ &\stackrel{(a)}{=} 1 - \Pr\left(|h_{S_bR}|^2 \geq \frac{\gamma_{\text{th}}}{\Phi}\right) \Pr\left(\frac{\gamma_{\text{BRD}_c}}{\tilde{\mu}} \geq \frac{\gamma_{\text{th}}}{\Phi}\right) \\ &= 1 - \bar{F}_{|h_{S_bR}|^2} \left( \frac{\gamma_{\text{th}}}{\Phi} \right) \bar{F}_{\gamma_{\text{BRD}_c}} \left( \frac{\tilde{\mu} \gamma_{\text{th}}}{\Phi} \right) \\ &\stackrel{(b)}{=} 1 + 2 \sum_{l_n, n \in (1, \dots, N)} \sum_{\substack{i_m, n_m, \\ m \in (1, \dots, M)}} \frac{(K_{n_t})^{i_t} (a_{n_t})^{j_t}}{i_t! j_t!} \left( \frac{\gamma_{\text{th}}}{\Phi} \right)^{\sum_{t=1}^m j_t} \\ &\quad \times \sqrt{\sum_{t=1}^n \frac{\tilde{\mu} \gamma_{\text{th}}}{\Phi \lambda_{BR} \lambda_{RD_t}}} \exp \left( - \sum_{t=1}^m \left[ K_{n_t} + \frac{a_{n_t} \gamma_{\text{th}}}{\Phi} \right] \right) \\ &\quad \times \mathcal{K}_1 \left( 2 \sqrt{\sum_{t=1}^n \frac{\tilde{\mu} \gamma_{\text{th}}}{\Phi \lambda_{BR} \lambda_{RD_t}}} \right), \end{aligned} \quad (32)$$

where  $\bar{F}_X(x)$  is the complementary cumulative distribution function (CCDF) of RV  $X$ , (a) is attained by considering the independent property of  $|h_{S_bR}|^2$  and  $\gamma_{\text{BRD}_c}$  and (b) is derived by substituting the CDF of  $|h_{S_bR}|^2$  and  $\gamma_{\text{BRD}_c}$  into Theorem 1 and (27). This ends the proof.

#### APPENDIX V PROOF OF EQUATION (19)

Here, we derive the exact mathematical framework of the IP. Let us begin with the definition of IP given in (33), as

shown at the bottom of the next page, where the last equation is held by conditioning the random variable  $x = |h_{\text{BE}}|^2$ , and  $\Lambda_3$  is computed from

$$\begin{aligned} \Lambda_3(x) &= \Pr \left( \max \left( \frac{\mu \Psi \Phi |h_{S_bR}|^2 |h_{BR}|^2 |h_{RE}|^2}{\Phi |h_{S_bR}|^2 (\Psi x + 1) + \mu \Psi |h_{BR}|^2 |h_{RE}|^2}, \right. \right. \\ &\quad \left. \left. \frac{\Phi |h_{S_bE}|^2}{\Psi x + 1} \right) < \gamma_{\text{th}} \right) \stackrel{(a)}{=} 1 - \Lambda_4(x) \Lambda_5(x), \\ \Lambda_4(x) &= \Pr \left( |h_{S_bE}|^2 < \frac{\gamma_{\text{th}} (\Psi x + 1)}{\Phi} \right) \\ &= 1 - \sum_{l \geq 0} \sum_{p=0}^l \frac{(K_m)^l (a_m)^p}{l! p!} \left( \frac{\gamma_{\text{th}} (\Psi x + 1)}{\Phi} \right)^p \\ &\quad \times \exp \left( -K_m - a_m \frac{\gamma_{\text{th}} (\Psi x + 1)}{\Phi} \right), \end{aligned} \quad (34)$$

Here, we obtain (a) due to the independent property of the two terms after conditioning RV  $x = |h_{\text{BE}}|^2$ . The last equation is immediately held by substituting the CDF of a non-central chi-square RV. To compute  $\Lambda_5$ , we first need the statistics of RV  $\gamma_{\text{BRE}} = |h_{BR}|^2 |h_{RE}|^2$  obtained from

$$\begin{aligned} F_{\gamma_{\text{BRE}}}(x) &= \Pr(\gamma_{\text{BRE}} = |h_{BR}|^2 |h_{RE}|^2 < x) \\ &= \Pr \left( |h_{RE}|^2 < \frac{x}{|h_{BR}|^2} \right) \\ &= \int_0^{+\infty} F_{|h_{RE}|^2} \left( \frac{x}{y} \right) f_{|h_{BR}|^2}(y) dy \\ &= \int_0^{+\infty} \frac{1}{\lambda_{BR}} \left( 1 - \exp \left( -\frac{x}{y \lambda_{RE}} \right) \right) \exp \left( -\frac{y}{\lambda_{BR}} \right) dy \\ &= 1 - 2 \sqrt{\frac{x}{\lambda_{BR} \lambda_{RE}}} \mathcal{K}_1 \left( 2 \sqrt{\frac{x}{\lambda_{BR} \lambda_{RE}}} \right), \\ f_{\gamma_{\text{BRE}}}(x) &= \frac{\partial F_{\gamma_{\text{BRE}}}(x)}{\partial x} = \frac{2}{\lambda_{BR} \lambda_{RE}} \mathcal{K}_0 \left( 2 \sqrt{\frac{x}{\lambda_{BR} \lambda_{RE}}} \right). \end{aligned} \quad (35)$$

Having the CDF and PDF of  $\gamma_{\text{BRE}}$ ,  $\Lambda_5$  is now compute by (36), as shown at the bottom of the next page. Here, we achieve (a) by employing the CDF of  $\gamma_{\text{BRE}}$  in (35) and substituting the CDF of  $|h_{S_bR}|^2$ . Undoubtedly, we cannot compute the integration in (36) as the closed-form expression in (30). Therefore, (33) is solely dependent on numerical computations. This ends the proof.

#### APPENDIX VI PROOF OF EQUATION (20)

Here, we derive the approximate mathematical framework of the IP. The IP of the approximated e2e SINR at E is given by (37), as shown at the bottom of the next page, where (a) is held by assuming that the channel gain from B to E in the second and third phase is uncorrelated; the CDF of  $\frac{\gamma_{\text{BRE}}}{|h_{\text{BE}}|^2}$  is

given as

$$\begin{aligned}
 & \frac{F_{\gamma_{\text{BRE}}}}{|h_{\text{BE}}|^2}(x) \\
 &= \int_0^{+\infty} F_{\gamma_{\text{BRE}}}(yx) f_{|h_{\text{BE}}|^2}(y) dy = \int_0^{+\infty} \frac{1}{\lambda_{\text{BE}}} \\
 & \times \left( 1 - 2\sqrt{\frac{xy}{\lambda_{\text{BR}}\lambda_{\text{RE}}}} K_1\left(2\sqrt{\frac{xy}{\lambda_{\text{BR}}\lambda_{\text{RE}}}}\right) \exp\left(-\frac{y}{\lambda_{\text{BE}}}\right) dy \right. \\
 & \quad \left. \stackrel{(a)}{=} 1 - \frac{4}{\lambda_{\text{BE}}} \sqrt{\frac{x}{\lambda_{\text{BR}}\lambda_{\text{RE}}}} \int_0^{+\infty} t^2 K_1\left(2\sqrt{\frac{xt}{\lambda_{\text{BR}}\lambda_{\text{RE}}}}\right) dt \right. \\
 & \quad \left. \times \exp\left(-\frac{t^2}{\lambda_{\text{BE}}}\right) \stackrel{(b)}{=} 1 - \exp\left(-\frac{\lambda_{\text{BE}}x}{2\lambda_{\text{BR}}\lambda_{\text{RE}}}\right) W_{-1, \frac{1}{2}}\left(\frac{\lambda_{\text{BE}}x}{\lambda_{\text{BR}}\lambda_{\text{RE}}}\right) \right) \quad (38)
 \end{aligned}$$

where (a) is held by changing variable  $y = t^2$ , (b) is obtained with the aid of [30, Eq. 6.631.3], and  $W_{a,b}(z)$  is the WhittakerW function. Finally, substituting (38) into (37),

$$\begin{aligned}
 \text{IP} &= \Pr(\gamma_{\text{E}} \geq \gamma_{\text{th}}) = \Pr\left(\max\left(\frac{\Phi|h_{\text{S}_b\text{E}}|^2}{\Psi|h_{\text{BE}}|^2 + 1}, \frac{\mu\Psi\Phi|h_{\text{S}_b\text{R}}|^2|h_{\text{BR}}|^2|h_{\text{RE}}|^2}{\Phi\Psi|h_{\text{S}_b\text{R}}|^2|h_{\text{BE}}|^2 + \mu\Psi|h_{\text{BR}}|^2|h_{\text{RE}}|^2 + \Phi|h_{\text{S}_b\text{R}}|^2}\right) \geq \gamma_{\text{th}}\right) \\
 &= 1 - \int_0^{+\infty} \underbrace{\Pr\left(\max\left(\frac{\Phi|h_{\text{S}_b\text{E}}|^2}{\Psi x + 1}, \frac{\mu\Psi\Phi|h_{\text{S}_b\text{R}}|^2|h_{\text{BR}}|^2|h_{\text{RE}}|^2}{\Phi|h_{\text{S}_b\text{R}}|^2(\Psi x + 1) + \mu\Psi|h_{\text{BR}}|^2|h_{\text{RE}}|^2}\right) < \gamma_{\text{th}}\right)}_{\Lambda_3(x)} f_{|h_{\text{BE}}|^2}(x) dx \quad (33)
 \end{aligned}$$

$\Lambda_5(x)$

$$\begin{aligned}
 &= \Pr\left(\frac{\mu\Psi\Phi|h_{\text{S}_b\text{R}}|^2\gamma_{\text{BRE}}}{\Phi|h_{\text{S}_b\text{R}}|^2(\Psi x + 1) + \mu\Psi\gamma_{\text{BRE}}} < \gamma_{\text{th}}\right) = \int_{\frac{\gamma_{\text{th}}(\Psi x + 1)}{\mu\Psi}}^{\infty} F_{|h_{\text{S}_b\text{R}}|^2}\left(\frac{\mu\Psi\gamma_{\text{th}}y}{\Phi(\mu\Psi y - \gamma_{\text{th}}(\Psi x + 1))}\right) f_{\gamma_{\text{BRE}}}(y) dy \\
 &+ \int_{y=0}^{\frac{\gamma_{\text{th}}(\Psi x + 1)}{\mu\Psi}} f_{\gamma_{\text{BRE}}}(y) dy \stackrel{(a)}{=} \left[1 - 2\sqrt{\frac{\gamma_{\text{th}}(\Psi x + 1)}{\mu\Psi\lambda_{\text{BR}}\lambda_{\text{RE}}}} K_1\left(2\sqrt{\frac{\gamma_{\text{th}}(\Psi x + 1)}{\mu\Psi\lambda_{\text{BR}}\lambda_{\text{RE}}}}\right)\right] + \frac{2}{\lambda_{\text{BR}}\lambda_{\text{RE}}} \int_{\frac{\gamma_{\text{th}}(\Psi x + 1)}{\mu\Psi}}^{\infty} \mathcal{K}_0\left(2\sqrt{\frac{y}{\lambda_{\text{BR}}\lambda_{\text{RE}}}}\right) \\
 &\times \left(1 - \sum_{\substack{i_m, n_m, \\ m \in (1, \dots, M)}} \frac{(K_{n_t})^{i_t} (a_{n_t})^{j_t}}{i_t! j_t!} \left(\frac{\mu\Psi\gamma_{\text{th}}y}{\Phi(\mu\Psi y - \gamma_{\text{th}}(\Psi x + 1))}\right)^{\sum_{t=1}^m j_t} \exp\left(-\sum_{t=1}^m \left(K_{n_t} + \frac{a_{n_t}\mu\Psi\gamma_{\text{th}}y}{\Phi(\mu\Psi y - \gamma_{\text{th}}(\Psi x + 1))}\right)\right) dy \right) \quad (36)
 \end{aligned}$$

$$\begin{aligned}
 \widetilde{\text{IP}} &= \Pr\left(\max\left(\frac{\Phi|h_{\text{S}_b\text{E}}|^2}{\Psi|h_{\text{BE}}|^2}, \frac{\mu\Phi|h_{\text{S}_b\text{R}}|^2|h_{\text{BR}}|^2|h_{\text{RE}}|^2}{\Phi|h_{\text{S}_b\text{R}}|^2|h_{\text{BE}}|^2 + \mu|h_{\text{BR}}|^2|h_{\text{RE}}|^2}\right) \geq \gamma_{\text{th}}\right) \\
 &= \Pr\left(\max\left(\frac{\Phi|h_{\text{S}_b\text{E}}|^2}{\Psi|h_{\text{BE}}|^2}, \min\left(\Phi|h_{\text{S}_b\text{R}}|^2, \frac{\mu\gamma_{\text{BRE}}}{|h_{\text{BE}}|^2}\right)\right) \geq \gamma_{\text{th}}\right) \\
 &\stackrel{(a)}{\approx} 1 - \Pr\left(|h_{\text{S}_b\text{E}}|^2 < \frac{\gamma_{\text{th}}\Psi|h_{\text{BE}}|^2}{\Phi}\right) \Pr\left(\min\left(\Phi|h_{\text{S}_b\text{R}}|^2, \frac{\mu\gamma_{\text{BRE}}}{|h_{\text{BE}}|^2}\right) < \gamma_{\text{th}}\right) = 1 - \Lambda_6\Lambda_7 \\
 \Lambda_6 &= \Pr\left(|h_{\text{S}_b\text{E}}|^2 < \frac{\gamma_{\text{th}}\Psi|h_{\text{BE}}|^2}{\Phi}\right) = \int_0^{+\infty} F_{|h_{\text{S}_b\text{E}}|^2}\left(\frac{\gamma_{\text{th}}\Psi x}{\Phi}\right) f_{|h_{\text{BE}}|^2}(x) dx \\
 &= 1 - \sum_{l \geq 0} \sum_{p=0}^l \frac{1}{\lambda_{\text{BE}}} \frac{(K_m)^l (a_m)^p}{l! p!} \int_0^{+\infty} \left(\frac{\gamma_{\text{th}}\Psi x}{\Phi}\right)^p \exp\left(-K_m - x \frac{a_m \gamma_{\text{th}}\Psi}{\Phi} - \frac{x}{\lambda_{\text{BE}}}\right) dx \\
 &= 1 - \sum_{l \geq 0} \sum_{p=0}^l \frac{\exp(-K_m)}{\lambda_{\text{BE}}} \frac{(K_m)^l (a_m)^p}{l!} \left(\frac{\gamma_{\text{th}}\Psi}{\Phi}\right)^p \left(\frac{a_m \gamma_{\text{th}}\Psi}{\Phi} + \frac{1}{\lambda_{\text{BE}}}\right)^{-1-p} \\
 \Lambda_7 &= \Pr\left(\min\left(\Phi|h_{\text{S}_b\text{R}}|^2, \frac{\mu\gamma_{\text{BRE}}}{|h_{\text{BE}}|^2}\right) < \gamma_{\text{th}}\right) = 1 - \Pr\left(\Phi|h_{\text{S}_b\text{R}}|^2 \geq \gamma_{\text{th}}\right) \Pr\left(\frac{\mu\gamma_{\text{BRE}}}{|h_{\text{BE}}|^2} \geq \gamma_{\text{th}}\right) \\
 &= 1 - \bar{F}_{\frac{\gamma_{\text{BRE}}}{\gamma_{\text{BE}}}}\left(\frac{\gamma_{\text{th}}}{\mu}\right) \left(\sum_{l \geq 0} \sum_{p=0}^l \frac{(K_m)^l (a_m)^p}{l! p!} \left(\frac{\gamma_{\text{th}}}{\Phi}\right)^p \exp\left(-K_m - a_m \frac{\gamma_{\text{th}}}{\Phi}\right)\right) \quad (37)
 \end{aligned}$$

we obtain the approximate closed-form expression of the IP given in (39).

$$\begin{aligned} \text{IP} &= 1 - \left[ 1 - \sum_{l \geq 0} \sum_{p=0}^l \frac{\exp(-K_m) (K_m)^l (a_m)^p}{\lambda_{BE} l!} \left( \frac{\gamma_{th} \Psi}{\Phi} \right)^p \right. \\ &\quad \times \left( \frac{a_m \gamma_{th} \Psi}{\Phi} + \frac{1}{\lambda_{BE}} \right)^{-1-p} \left[ 1 - \exp \left( \frac{\lambda_{BE} \gamma_{th}}{2 \lambda_{BR} \lambda_{RE} \mu} \right) \right. \\ &\quad \times \left( \sum_{l \geq 0} \sum_{p=0}^l \frac{(K_m)^l (a_m)^p}{l! p!} \left( \frac{\gamma_{th}}{\Phi} \right)^p \exp \left( -K_m - a_m \frac{\gamma_{th}}{\Phi} \right) \right. \\ &\quad \times \left. W_{-1, \frac{1}{2}} \left( \frac{\lambda_{BE} \gamma_{th}}{\lambda_{BR} \lambda_{RE} \mu} \right) \right] \left. \right]. \quad (39) \end{aligned}$$

This ends the proof.

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